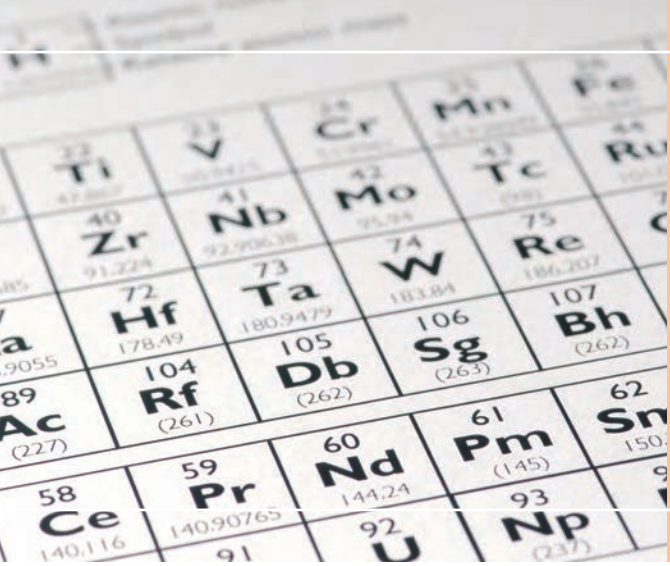
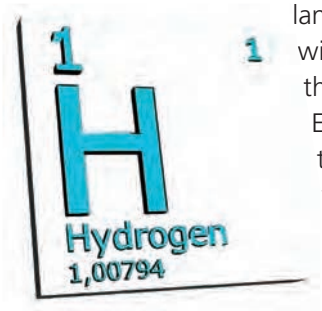


Topic 2

Microscopic World I

Chemistry in Context

Common language in the chemistry world Which language do you think is the most widely used in the world? Before the 21st century, the answer was English. It now becomes hard to tell as more native speakers from various countries start to use Putonghua to communicate as well. In the world of chemistry, there is a common language used by chemists all over the world. Chemists use chemical symbols, formulae and equations to communicate with one another. For example, a chemist uses 'H' for hydrogen and 'He' for helium. These chemical symbols are the internationally agreed codes for all chemical elements ever discovered, and many of them are derived from the English or Latin names of the elements. The International Union of Pure and Applied Chemistry (IUPAC) sets the global standards for names and symbols of chemical elements and compounds.



Do you recall?

1 What is the chemical symbol for water?
H_2O

2 Pure water can be obtained by distillation in a laboratory. Suggest a test to show that the distillate is water.
The distillate turns blue cobalt(II) chloride paper pink.

3 The hydrogen atoms and oxygen atom in a water molecule are held strongly together. How can we break the strong attractive force between the atoms?
Electrolysis

In this topic, you are going to acquire knowledge of some basic chemical principles through studying the structures, bonding and properties of various substances.

Unit

5 Atomic structure

Unit Key Concepts

- Atomic number and mass number
- Isotopes
- Relative atomic mass
- Electronic arrangement of atoms

Pre-study

A pre-study worksheet on this unit is available for download from *Jing Kung Chemistry Website*.

5.1 What is an element made of?

Imagine that we cut a piece of copper into two halves, and then cut each smaller piece into two halves again. If this process were repeated continuously (Fig. 5.1), we would finally get to the stage where the individual pieces would be **atoms** of copper.

It is possible to break these atoms into even smaller pieces (i.e. subatomic particles), but those pieces would no longer be copper.

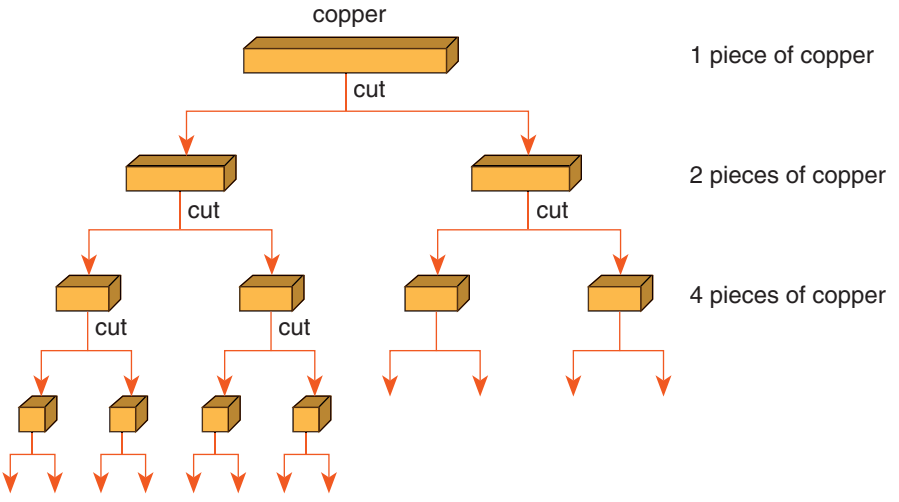
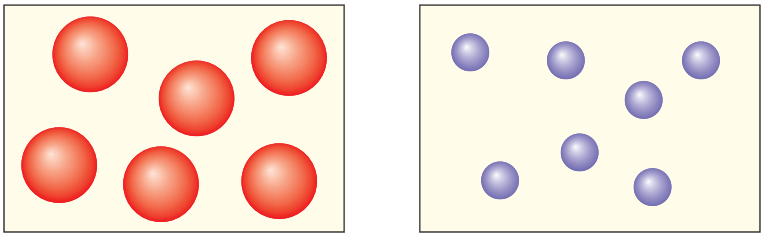


Fig. 5.1 Cutting a piece of copper into halves

All **elements** are made of atoms. The atoms of different elements are different (Fig. 5.2).



(a) Element A (b) Element B

Fig. 5.2 Atoms of different elements are different

atom 原子 element 元素

5.2 Symbols for elements

Chemists use **symbols** to represent elements. These symbols are a form of chemical shorthand. They fall into three categories.

Some symbols are simply the first letter of the names of the elements.

Element	Carbon	Fluorine	Hydrogen	Nitrogen	Oxygen	Phosphorus	Sulphur
Symbol	C	F	H	N	O	P	S

Where there are several elements beginning with the same letter, a second letter is used, frequently the second letter of the name of the element.

Element	Aluminium	Argon	Calcium	Chlorine	Magnesium	Silicon
Symbol	Al	Ar	Ca	Cl	Mg	Si

Some symbols are derived from the Latin names of the elements.

Element	Copper	Gold	Iron	Potassium	Sodium
Symbol	Cu	Au	Fe	K	Na
Latin name	Cuprum	Aurum	Ferrum	Kalium	Natrium

Fig. 5.3 shows some common elements and their symbols.

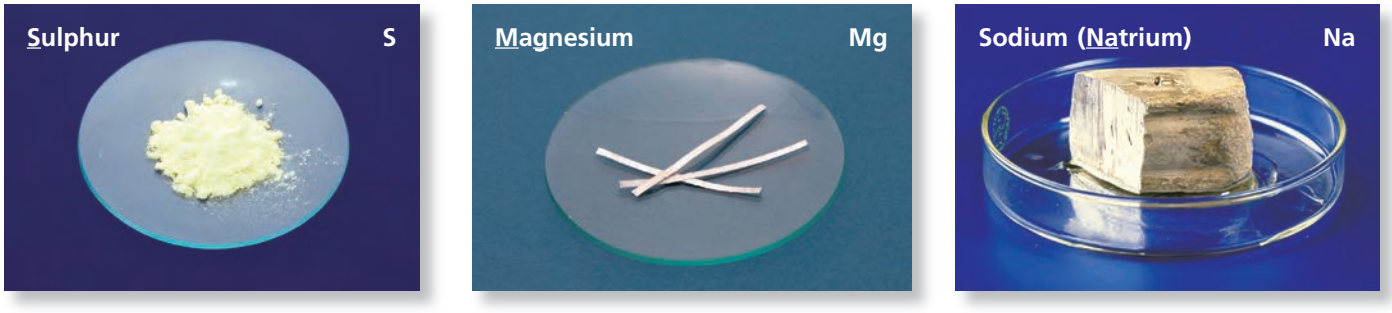


Fig. 5.3 Some common elements and their symbols

symbol 符號

Practice 5.1

Complete the following table.

Element	Aluminium	Calcium	Hydrogen	Iron	Sodium	Chlorine	Copper	Nitrogen	Oxygen	Sulphur
Symbol	Al	Ca	H	Fe	Na	Cl	Cu	N	O	S

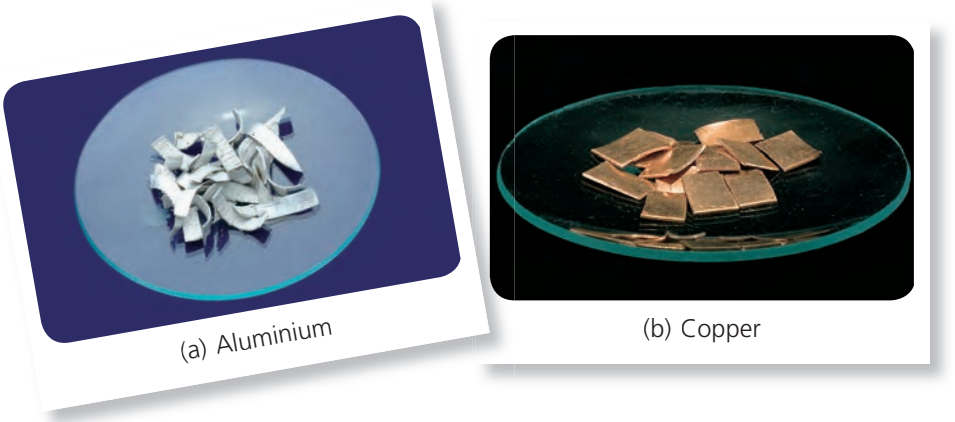
5.3 How to classify elements?

5.1
Investigating the physical properties of metals and non-metals.

Metals and non-metals

We can classify elements in many different ways. We can use the properties of an element to classify it as a **metal** or a **non-metal**. N1 Figs. 5.4 and 5.5 show some metals and non-metals.

N1 About three fourths of the elements are metals. All the metals except mercury are solids at room temperature and pressure.



(a) Aluminium (b) Copper

Fig. 5.4 Aluminium and copper are metals



(a) Sulphur (b) Chlorine

Fig. 5.5 Sulphur and chlorine are non-metals

metal 金屬 non-metal 非金屬

Worksheet
A worksheet on this activity is available for download from *Jing Kung Chemistry Website*.

Discussion

The following table lists the physical properties of some elements. Aluminium, copper and gold are metals while the others are non-metals.

Element		Melting point (°C)	Boiling point (°C)	Electrical conductivity (***** = best, * = worst, — = does not conduct electricity)	Heat conductivity (***** = best, * = worst)
Metals	Aluminium	660	2 467	****	****
	Copper	1 083	2 567	*****	*****
	Gold	1 063	3 080	*****	*****
Non-metals	Chlorine	–101	–34	—	*
	Oxygen	–218	–183	—	*
	Sulphur	119	445	—	*

Based on the information in the table, suggest the differences in the physical properties of metals and non-metals. Refer to page T18.



Fig. 5.6 Mercury has a low melting point; it is a liquid at room temperature and pressure

Metals and non-metals have quite different properties. Most of the elements are metals. Among the 112 known elements, only 17 are non-metals.

Fig. 5.7 shows the typical properties of metals.

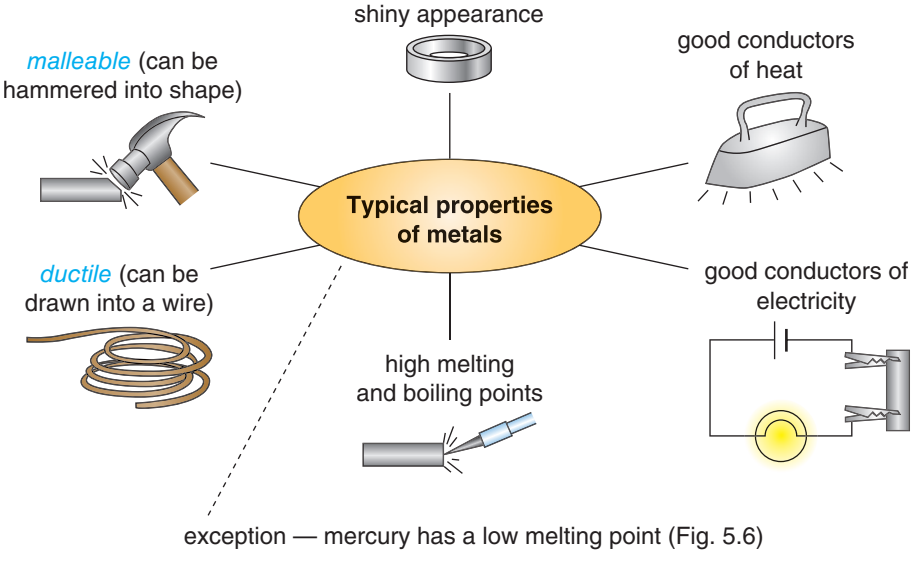


Fig. 5.7 Typical properties of metals

malleable 可展的　　ductile 可延的

Fig. 5.8 shows the typical properties of non-metals.

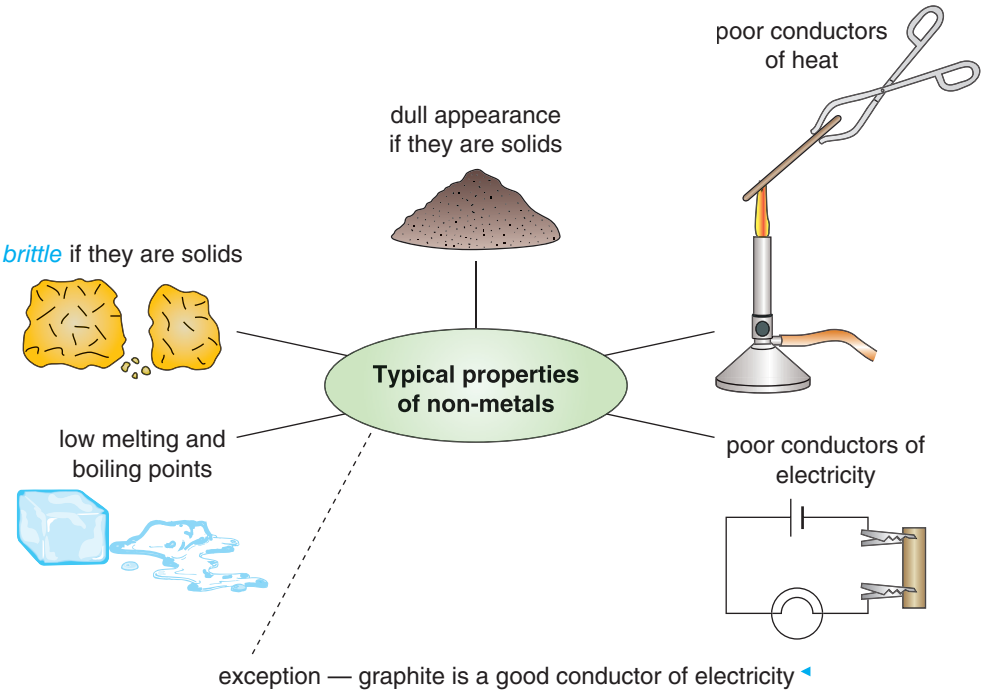


Fig. 5.8 Typical properties of non-metals

We will discuss the electrical conductivity of graphite in Unit 9.

Metalloids

Metalloids are also known as semi-metals.

Study the properties of silicon as shown in Table 5.1. Notice that silicon has some properties similar to those of metals and some properties similar to those of non-metals. Silicon belongs to a group of elements called **metalloids**. Other examples of metalloids are boron and germanium.

Table 5.1

Some properties of silicon	
Property	**Silicon (a metalloid)**
State at room temperature and pressure	solid
Melting and boiling point	high
Appearance	grey and shiny crystals, or brown powder
Electrical conductivity	crystalline form conducts electricity, brown powder form does not conduct electricity
Effect of bending and hammering	brittle

brittle 易碎的　　metalloid 類金屬　　state 狀態

These metalloids are very useful because semi-conductors are needed in making integrated circuits in the electronics industry.

Practice 5.2

Study the following descriptions of three elements. Classify each as a metal, non-metal or metalloid. Explain your choice in each case.

Element	Description
X	a yellow solid that melts at 119 °C; both the solid and liquid forms do not conduct electricity
Y	a shiny solid which can be bent or hammered into shapes easily
Z	a shiny brittle solid which can conduct electricity

Practice 5.2

X is a non-metal.
It has a low melting point and does not conduct electricity.
Y is a metal.
It is shiny and can be bent or hammered into shapes.
Z is a metalloid.
It has the property of a non-metal (a brittle solid) and the property of a metal (can conduct electricity).

5.4 Basic structure of an atom

Every atom has nearly all of its mass concentrated in a tiny region in the centre of the atom called the **nucleus**. The nucleus contains two types of particles[1]: **protons** and **neutrons**.

[1] Modern experiments show that an atom consists of many different types of particles. What we consider here is only a simple model.

The nucleus is surrounded by a cloud of rapidly moving **electrons**. This electron cloud constitutes the bulk of the volume of the atom (Fig. 5.9).

Animation
Electrons move around the nucleus

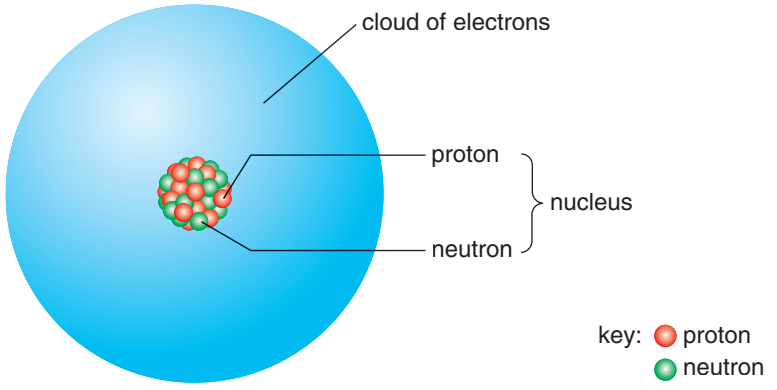


Fig. 5.9 A model for the structure of an atom

nucleus 原子核　proton 質子　neutron 中子　electron 電子

Protons, neutrons and electrons are called **subatomic particles**. These particles are very small and have very little mass.

Protons and neutrons have almost the same mass. Electrons are very light. The mass of one proton is approximately equal to that of 1 840 electrons.

A proton has a small positive electrical charge. An electron has an equal but opposite (negative) charge. A neutron has no charge. Table 5.2 shows the data of the three types of subatomic particles.

To help you remember the charge on the subatomic particles, note that:

- **p**rotons are **p**ositive;
- **n**eutrons are **n**eutral;
- so that electrons must be negative.

Table 5.2

Data of subatomic particles

Name	Symbol	Position in atom	Relative charge	Relative mass
Proton	p	inside the nucleus	+1	1
Neutron	n	inside the nucleus	0	1
Electron	e^-	moving around the nucleus	–1	$\frac{1}{1\,840}$

As an atom is electrically neutral, the number of electrons must equal the number of protons.

✓ In a neutral atom,
number of electrons = number of protons

So, an oxygen atom having 8 protons must also have 8 electrons.

5.5 Atomic number

The **atomic number** (symbol: Z) of an element is the number of protons in an atom of that element.

✓ Atomic number of an element
= number of protons in an atom of that element

subatomic particle 次原子粒子　　atomic number 原子序

The atomic number tells us the following:

- what the element is — every element has a unique atomic number; N2
- the number of electrons in a neutral atom — the number of electrons must equal the number of protons.

N2 Examination questions often ask students to identify elements based on their atomic numbers.

A hydrogen atom has 1 proton in its nucleus. Thus, the atomic number of hydrogen is 1. A boron atom has 5 protons in its nucleus. Thus, the atomic number of boron is 5 (Fig. 5.10).

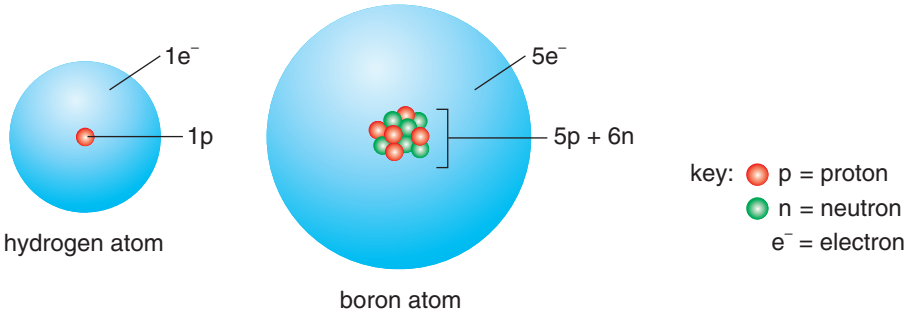


Fig. 5.10 Structures of a hydrogen atom and a boron atom (atomic number: hydrogen = 1, boron = 5)

5.6 Mass number

The **mass number** (symbol: A) of an atom is the sum of the numbers of protons and neutrons in the atom.

✓ Mass number = number of protons + number of neutrons

We can obtain the number of neutrons in an atom by subtracting the atomic number from the mass number.

✓ Number of neutrons = mass number – atomic number

mass number 質量數

Notice that a hydrogen atom has no neutron (applicable to protium only).

Table 5.3 lists the number of subatomic particles in atoms of the first 20 elements. For example, a lithium atom has 3 protons, 4 neutrons and 3 electrons. An oxygen atom has 8 protons, 8 neutrons and 8 electrons. N3

N3 Remind students that they need to memorize the number of protons and electrons, but NOT the number of neutrons, in atoms of the first 20 elements.

Table 5.3

Number of subatomic particles in atoms of the first 20 elements

Atom	Symbol	Number of protons (atomic number)	Number of neutrons	Number of electrons	Mass number
Hydrogen	H	1	0	1	1
Helium	He	2	2	2	4
Lithium	Li	3	4	3	7
Beryllium	Be	4	5	4	9
Boron	B	5	6	5	11
Carbon	C	6	6	6	12
Nitrogen	N	7	7	7	14
Oxygen	O	8	8	8	16
Fluorine	F	9	10	9	19
Neon	Ne	10	10	10	20
Sodium	Na	11	12	11	23
Magnesium	Mg	12	12	12	24
Aluminium	Al	13	14	13	27
Silicon	Si	14	14	14	28
Phosphorus	P	15	16	15	31
Sulphur	S	16	16	16	32
Chlorine	Cl	17	18	17	35
Argon	Ar	18	22	18	40
Potassium	K	19	20	19	39
Calcium	Ca	20	20	20	40

Chemical shorthand

We can use a shorthand convention to show the atomic number and mass number of an atom. Fig. 5.11 shows the shorthand convention for an atom of sodium, which has 11 protons, 12 neutrons and 11 electrons. The numbers of protons and neutrons add up to a mass number of 23. N4

N4 Examination questions often ask students to give the number of subatomic particles in an atom of an unfamiliar element (e.g. ^{90}Sr).

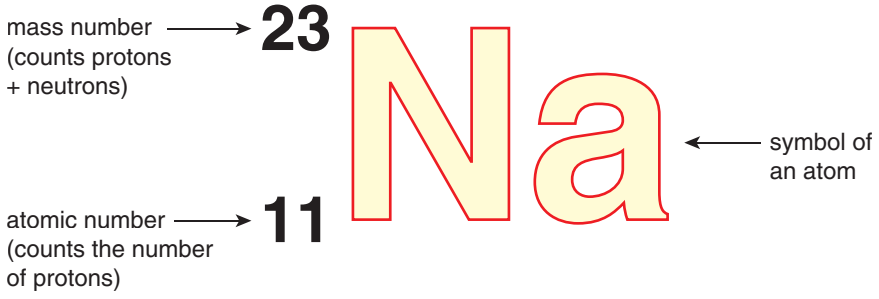


Fig. 5.11 The shorthand convention for showing the atomic number and mass number of a sodium atom

Example 5.1

Q A phosphorus atom contains 16 neutrons. Write a shorthand convention to show the atomic number and mass number of this atom.

A Firstly, find out the atomic number of phosphorus. The atomic number of phosphorus is 15 and its symbol is P.

Mass number
= number of protons + number of neutrons
= 15 + 16
= 31

∴ the shorthand convention for the phosphorus atom is $^{31}_{15}$P.

Practice 5.3

1 Work out the mass number, atomic number and the number of subatomic particles of each atom below.

Atom	Mass number	Atomic number	Number of		
			protons	neutrons	electrons
$^{56}_{26}$Fe	56	26	26	30	26
$^{93}_{41}$Nb	93	41	41	52	41
$^{200}_{80}$Hg	200	80	80	120	80

2 Consider the following two atoms: ^{131}I and ^{131}Xe.

In terms of the numbers of protons, neutrons and electrons, suggest how ^{131}I and ^{131}Xe are different from each other.

DSE 2012 Paper 1A Q15

Practice 5.3
2 • They have different numbers of protons / electrons.
They are atoms of different elements.
• They have different numbers of neutrons.
They have the same mass number (i.e. the numbers of protons + neutrons are the same). However, they have different number of protons.

Reference Website
Refer to a video clip on isotopes:
http://www.youtube.com/watch?v=eF3ujFJ3hkk
(accessed July 2014)

5.7 Isotopes

All atoms of the same element have the same number of protons in the nucleus. However, atoms of the same element do not necessarily have the same mass. Atoms of the same element with different masses have a different number of neutrons in the nucleus.

CE 2007 Paper 2 Q29
CE 2008 Paper 1 Q2(a)
DSE 2012 Paper 1B Q1(a)

✓ **Isotopes** are different atoms of an element which have the same number of protons but a different number of neutrons.

Consider the two isotopes of chlorine (Fig. 5.12). Both of them have 17 protons. The difference between them is that one has 18 neutrons while the other has 20 neutrons.

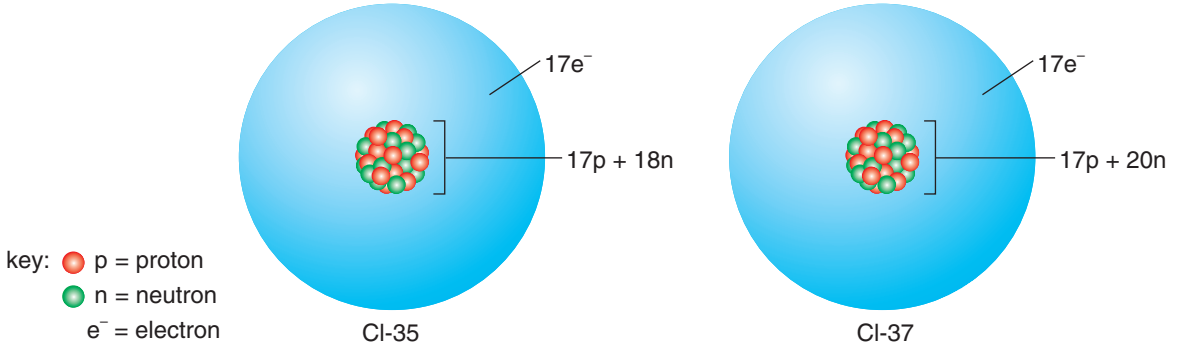


Fig. 5.12 The two isotopes of chlorine

We can represent an isotope by writing the symbol of the atom followed by its mass number. These two chlorine isotopes are Cl-35 and Cl-37.

N5 Refer to page T18.

Most elements have more than one isotope. Table 5.4 gives a N5 list of some well-known isotopes. The three isotopes of hydrogen are given special names.

Table 5.4

Isotopes of some elements

Element	Name of isotope	Symbol	Mass number	Number of		
				protons	neutrons	electrons
Hydrogen	*Protium*	$^{1}_{1}H$	1	1	0	1
	Deuterium	$^{2}_{1}H$	2	1	1	1
	Tritium	$^{3}_{1}H$	3	1	2	1
Carbon	Carbon-12	$^{12}_{6}C$	12	6	6	6
	Carbon-13	$^{13}_{6}C$	13	6	7	6
	Carbon-14	$^{14}_{6}C$	14	6	8	6

isotope 同位素　　protium 氕（音：撇）　　deuterium 氘（音：刀）　　tritium 氚（音：川）

N6 Isotopes of the same element have the same electronic arrangement and hence the same chemical properties. They cannot be separated by chemical means.
e.g. $^{10}BCl_3$ and $^{11}BCl_3$ have the same chemical properties.

Isotopes of the same element have identical chemical properties N6 because they have the same number of electrons. It is the number of electrons in its atom that determines the chemical properties of an element. However, their physical properties differ slightly. For example, the boiling point of $^{37}_{17}Cl_2$ is higher than that of $^{35}_{17}Cl_2$.

Practice 5.4

1 Lead exists as a mixture of isotopes: $^{206}_{82}Pb$, $^{207}_{82}Pb$ and $^{208}_{82}Pb$.

Complete the table below to show the similarities and differences among these isotopes.

Isotope	Mass number	Number of		
		protons	neutrons	electrons
$^{206}_{82}Pb$	206	82	124	82
$^{207}_{82}Pb$	207	82	125	82
$^{208}_{82}Pb$	208	82	126	82

2 Consider the following information about atoms X, Y and Z:

Atom	Number of		
	protons	neutrons	electrons
X	8	8	8
Y	8	10	8
Z	10	10	10

Which of the above atoms are isotopes? Explain your choice.

X and Y are isotopes.
They have the same number of protons but a different number of neutrons.

5.8 Relative masses of atoms and the carbon-12 scale

Atoms are so light that it is difficult to weigh them in standard units of mass, like grams or kilograms. Instead of finding the masses of atoms directly, we compare the masses of different atoms in a scale based on the mass of an atom of the $^{12}_{6}C$ isotope.

One carbon-12 atom contains six protons and six neutrons and has a mass number of 12. Chemists have defined the carbon-12 atom as having a mass of exactly 12.00.

Relative isotopic mass

✓ The **relative isotopic mass** of a particular isotope of an element is the mass of one atom of the isotope compared with $\frac{1}{12}$ of the mass of one carbon-12 atom.

i.e. relative isotopic mass of an isotope

$$= \frac{\text{mass of one atom of the isotope}}{\frac{1}{12} \times \text{mass of one carbon-12 atom}}$$

All the atoms of an isotope are identical. They all have the same mass. N7

The mass of a hydrogen atom ($^{1}_{1}H$) is equal to $\frac{1}{12}$ of the mass of a carbon-12 atom, so its relative isotopic mass is 1. A magnesium atom ($^{24}_{12}Mg$) is twice as heavy as a carbon-12 atom, so its relative isotopic mass is 24 (Fig. 5.13).

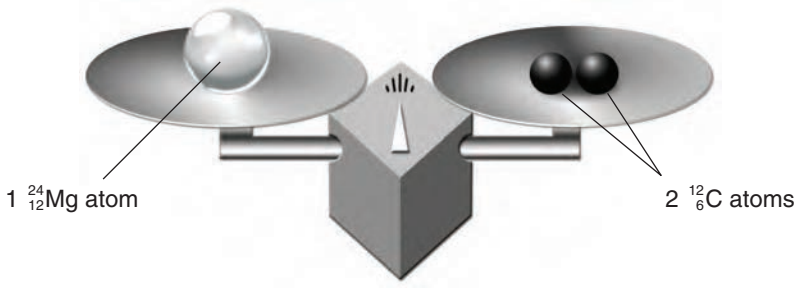


Fig. 5.13 A magnesium atom ($^{24}_{12}Mg$) is twice as heavy as a carbon-12 ($^{12}_{6}C$) atom

Do you know

Why carbon-12?

Before 1961, oxygen was used as the standard for atomic masses. Oxygen was used rather than the lightest atom, hydrogen, because oxygen combines with many substances to form oxides.

Unfortunately, chemists had based their scale on naturally occurring oxygen, which has a mixture of the isotopes oxygen-16, oxygen-17 and oxygen-18, whereas physicists had chosen the single isotope oxygen-16 on which to base atomic masses. This gave two different sets of relative atomic masses!

In 1961, chemists and physicists finally compromised that all atomic masses would be compared with the carbon-12 isotope. This scale required only minimal changes to the values that had been used for relative atomic masses.

N7 Different isotopes of an element have different masses.

relative isotopic mass 相對同位素質量

On this scale, the relative masses of a proton and a neutron are both very near to 1 while that of an electron is near to 0. Thus, the relative isotopic mass of an isotope is roughly equal to its mass number. So, for oxygen-16 ($^{16}_{8}O$), the relative isotopic mass is 16.

Relative atomic mass

N8 Examination questions often ask students to calculate the relative atomic mass of an element based on the relative isotopic mass and relative abundance of each isotope of the element.

N9 Refer to page T19.

Most elements contain a mixture of isotopes, each in a different amount and with a different mass. The contribution made by an isotope to the **relative atomic mass** of an element depends on: N8 N9

- the **relative abundance** of the isotope; and
- the relative isotopic mass of the isotope.

✓ The relative atomic mass◂ of an element is the weighted average relative isotopic mass of all the naturally occurring isotopes of that element compared with $\frac{1}{12}$ of the mass of one carbon-12 atom.

◂ Relative atomic mass carries no unit because it is a ratio.

CE 2006 Paper 2 Q14
DSE 2012 Paper 1B Q1(b)
ASL 2009 Paper 1 Q2(a)
AL 2009 Paper 1 Q3(b)(i)

Example 5.2

Q Chlorine has two isotopes: $^{35}_{17}Cl$ and $^{37}_{17}Cl$. The relative abundance of the isotopes in nature is shown below:

Isotope	Relative isotopic mass	Relative abundance (%)
$^{35}_{17}Cl$	35	75
$^{37}_{17}Cl$	37	25

Calculate the relative atomic mass of chlorine.

A Relative atomic mass of chlorine

= weighted average◂ relative isotopic mass of the isotopes

$$= \frac{35 \times 75 + 37 \times 25}{100}$$

= 35.5◂

∴ the relative atomic mass of chlorine is 35.5.

◂ Notice the effect of taking 'weighted average'. The average of 35 and 37 is 36. Our answer of 35.5 reflects that there are more of the lighter isotope of chlorine.

◂ The value of 35.5 is not the relative mass of any one chlorine atom. It is the weighted average relative isotopic mass of all the naturally occurring isotopes of chlorine.

relative atomic mass 相對原子質量　　relative abundance 相對豐度

N10 Examination questions may give the relative atomic mass of an element and ask students to calculate the relative abundance of the isotopes of the element.

CE 2008 Paper 1 Q2(b)
CE 2010 Paper 2 Q8

Example 5.3

Q Magnesium has three isotopes: ^{24}Mg, ^{25}Mg and ^{26}Mg. The relative abundance of the ^{26}Mg isotope is 10%. The relative atomic mass of Mg is 24.3. What is the relative abundance of the ^{24}Mg isotope? N10

A Let the relative abundance of ^{24}Mg isotope and ^{25}Mg isotope be y% and $(90 - y)$% respectively.

Relative atomic mass of magnesium = weighted average relative isotopic mass of the isotopes

$$24.3 = \frac{24 \times y + 25 \times (90 - y) + 26 \times 10}{100}$$

$$2\,430 = 24y + 2\,250 - 25y + 260$$

$$y = 80$$

$\therefore$ the relative abundance of the ^{24}Mg isotope is 80%.

N11 Remind students that they do NOT need to memorize the relative atomic masses of elements. These will be given in examinations.

Table 5.5 lists the relative atomic masses of some common elements. N11

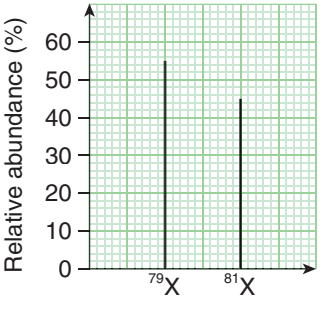
Table 5.5

Relative atomic masses of some common elements

Element	Symbol	Relative atomic mass	Element	Symbol	Relative atomic mass
Aluminium	Al	27.0	Hydrogen	H	1.0
Calcium	Ca	40.1	Magnesium	Mg	24.3
Chlorine	Cl	35.5	Oxygen	O	16.0
Copper	Cu	63.5	Potassium	K	39.1

Practice 5.5

1 Suppose that element X has two isotopes, ^{79}X and ^{81}X. The graph below shows the relative abundance of the two isotopes:



What is the relative atomic mass of X?

2 Lithium consists of two isotopes: ^{6}Li and ^{7}Li. The relative atomic mass of lithium is 6.9. Calculate the relative abundance of the isotopes.

1 Relative atomic mass of X $= \frac{79 \times 55 + 81 \times 45}{100}$

$= 79.9$

$\therefore$ the relative atomic mass of X is 79.9.

2 Let the relative abundance of ^{6}Li isotope and ^{7}Li isotope be y% and $(100 - y)$% respectively.

Relative atomic mass of Li $= 6.9 = \frac{6 \times y + 7 \times (100 - y)}{100}$

$690 = 6y + 700 - 7y$

$y = 10.0$

$\therefore$ the relative abundance of the ^{6}Li isotope is 10.0% and that of ^{7}Li isotope is 90.0%.

DSE 2013 Paper 1A Q2

The Bohr's model is a useful model of an atom. However, it is not the only model. Sometimes it is better to consider electrons as moving around the nucleus in the form of a cloud. We will discuss this later in this unit.

5.9 The arrangement of electrons in atoms

Electronic arrangement

Niels Bohr (Fig. 5.14) was a Danish scientist. In the early 20th century, he suggested that electrons move around the nucleus in circular **orbits** called **shells** (Fig. 5.15).

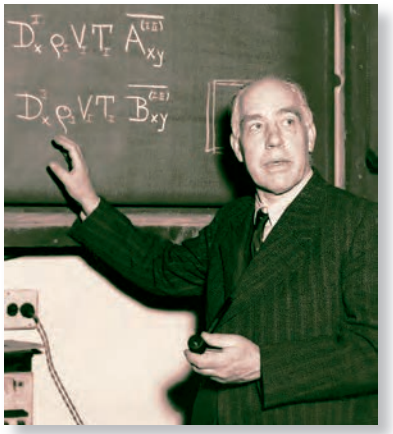


Fig. 5.14 Niels Bohr

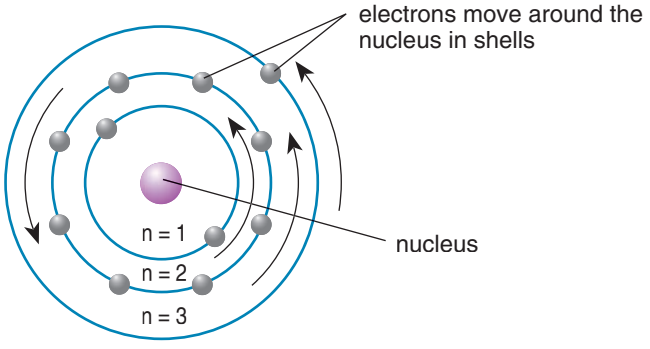


Fig. 5.15 Electrons move around the nucleus in shells

Each shell is given a number (n). The shell nearest to the nucleus is given the number of 1 (i.e. n = 1), the next shell is given the number of 2 (i.e. n = 2), etc.

Each shell can only hold a certain number of electrons. The maximum number of electrons a certain shell can hold is given by the expression of $2n^2$, where n is the shell number (Fig. 5.16).

Reference Website
Refer to a video clip on atomic structure:
http://www.bbc.co.uk/schools/gcsebitesize/science/add_ocr_gateway/periodic_table/atomstrucact.shtml
(accessed July 2014)

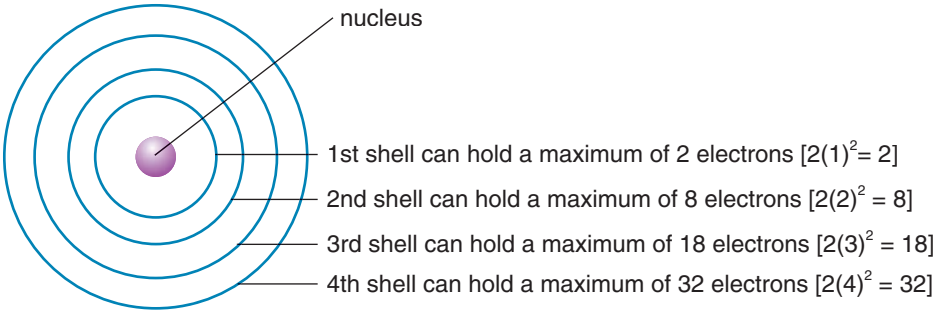


Fig. 5.16 The maximum number of electrons the first four shells can hold

orbit 軌道　　shell 電子層

An electron diagram is just a way of representing the electronic arrangement of an atom. In fact, the electrons do not stay in fixed positions and the shells are not circular in shape.

N12 Notice that the electrons (except those in the first shell) are drawn in pairs for easy counting. Remind students that they need to memorize the electronic arrangements of atoms of the first 20 simplest elements. Examination questions may ask students to identify elements based on the electronic arrangements of their atoms.

Animation
Electronic arrangements and electron diagrams of atoms

The way in which electrons are arranged in an atom is called its **electronic arrangement**. We can represent the electronic arrangement of an atom by an **electron diagram**. N12

Fig. 5.17 shows the electron diagrams of atoms of the two lightest elements, hydrogen and helium. A hydrogen atom has only 1 electron. The electron goes into the first shell. A hydrogen atom has an electronic arrangement of 1. A helium atom has 2 electrons. Both electrons go into the first shell and the first shell is full. A helium atom has an electronic arrangement of 2.



Fig. 5.17 Electron diagrams of a hydrogen atom and a helium atom

Filling electrons into shells of oxygen and sodium atoms

The following guidelines will apply when filling electrons into shells:

- electrons are filled into shells starting from the first shell which is closest to the nucleus;
- start filling the next shell when one shell is full.

An oxygen atom has 8 electrons. The first 2 electrons go into the first shell while the remaining 6 electrons go into the second shell. Thus, an oxygen atom has an electronic arrangement of 2,6 (Fig. 5.18a).

A sodium atom has 11 electrons. The first 10 electrons fill up the first and second shells while the last electron goes into the third shell. A sodium atom thus has an electronic arrangement of 2,8,1 (Fig. 5.18b).

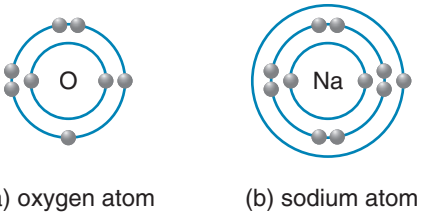


Fig. 5.18 Electron diagrams of an oxygen atom and a sodium atom

electronic arrangement 電子排佈　　electron diagram 電子圖

Electronic arrangements of atoms of the first 20 elements

Table 5.6 shows the electronic arrangements of atoms of the first 20 elements (atomic number from 1 to 20).

Table 5.6

Electronic arrangements of atoms of the first 20 elements

Atomic number	Element	Number of electrons in				Electronic arrangement
		1st shell	2nd shell	3rd shell	4th shell	
1	Hydrogen	1				1
2	Helium	2				2 (first shell is full)
3	Lithium	2	1			2,1
4	Beryllium	2	2			2,2
5	Boron	2	3			2,3
6	Carbon	2	4			2,4
7	Nitrogen	2	5			2,5
8	Oxygen	2	6			2,6
9	Fluorine	2	7			2,7
10	Neon	2	8			2,8 (second shell is full)
11	Sodium	2	8	1		2,8,1
12	Magnesium	2	8	2		2,8,2
13	Aluminium	2	8	3		2,8,3
14	Silicon	2	8	4		2,8,4
15	Phosphorus	2	8	5		2,8,5
16	Sulphur	2	8	6		2,8,6
17	Chlorine	2	8	7		2,8,7
18	Argon	2	8	8		2,8,8
19	Potassium	2	8	8	1	2,8,8,1
20	Calcium	2	8	8	2	2,8,8,2

The reason for this is too complicated and we will not explain this at this level. You will learn this at a higher level.

Notice the electronic arrangements of potassium and calcium atoms (Fig. 5.19). The third shell can hold 18 electrons. However, in the case of potassium or calcium atoms, only 8 electrons go into the third shell. The remaining electrons go into the fourth shell.

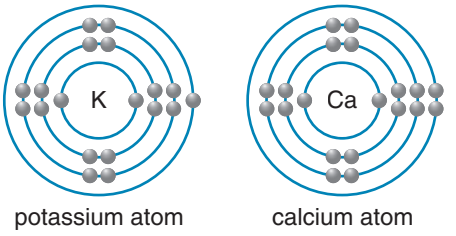


Fig. 5.19 Electron diagrams of a potassium atom and a calcium atom

Practice 5.6

1 Draw electron diagrams of the following atoms:

a) $^{11}_{5}B$

b) $^{20}_{10}Ne$

c) $^{35}_{17}Cl$

2 Complete the following table about the electronic arrangements of atoms of some elements.

Element	Atomic number	Electronic arrangement of atom
Aluminium	13	2,8,3
Fluorine	9	2,7
Magnesium	12	2,8,2

3 This question refers to the electronic arrangements of atoms of some elements.

Element	Electronic arrangement of atom
W	2,4
X	2,8,8
Y	2,8,8,2
Z	2,8,18,18,7

a) Which of the elements is carbon?
W

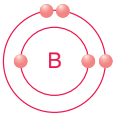
b) Name element Y.
Calcium

c) How many protons does element Z contain?
53

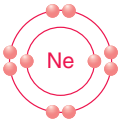
d) An atom of element A has one more electron than an atom of element X.

Draw an electron diagram of an atom of A.

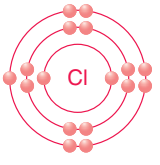
1 a)



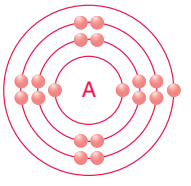
b)



c)



3 d)



5.10 Electrons and orbitals

So far, we have described electrons in atoms as occupying different shells at increasing distances from the nucleus.

But, suppose we could photograph the electron in a hydrogen atom at any given moment. The electron is moving at a high speed. The electron would occupy different positions if we took photographs at different moments. If we superimposed millions of such photographs, the resulting picture would resemble a cloud composed of a great number of dots, each dot representing one position of the electron at a particular instant (Fig. 5.20).

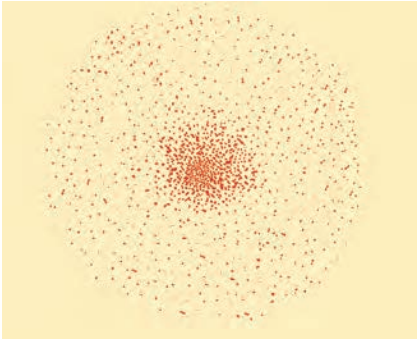


Fig. 5.20 The electron cloud in a hydrogen atom

If you wanted to be absolutely 100% sure of where the electron is, you would have to draw a sphere of the size of the Universe.

Thus, in the hydrogen atom, we can imagine the electron as an electron cloud. In theory, there is no sharp boundary to the electron cloud. But we can draw a sphere enclosing about 95% of the cloud (Fig. 5.21). Within the region enclosed by the sphere, there is 95% chance of finding the electron. The region in which there is this high probability of finding the electron is called an **orbital**.

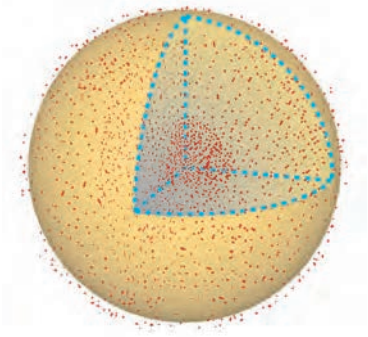


Fig. 5.21 The sphere enclosing 95% of the cloud defines an orbital (a portion of the sphere and electron cloud has been taken away for easy visualization)

orbital 軌態

Worksheet

Two related worksheets with answers are available for download from *Jing Kung Chemistry Website*. Worksheet ① consists of additional questions for promoting active reading. (Note: Questions in this worksheet also cover those in the textbook.) Worksheet ②, requiring students to indicate the relationship among various terms by drawing, may help the less able readers improve their reading skills.

Chemistry Magazine

How ideas of the atom developed

All chemistry depends on one big idea: that everything is made of atoms. But how did chemists find out about atoms?



Fig. 5.22 John Dalton

The atomic theory

In the early 1800s, John Dalton (Fig. 5.22) developed his atomic theory. This included the following predictions about atoms:

- Atoms are tiny particles that make up elements.
- Atoms cannot be divided.
- All atoms of a given element are the same.
- Atoms of one element are different from those of every other element.

Discovery of electrons and protons

In 1897, J. J. Thomson passed high voltage electricity through a gas in a tube at low pressure. He found that a stream of rays, called cathode rays, moved from the negative electrode to the positive electrode. Besides, cathode rays were also deflected strongly towards the positive plate (Fig. 5.23).

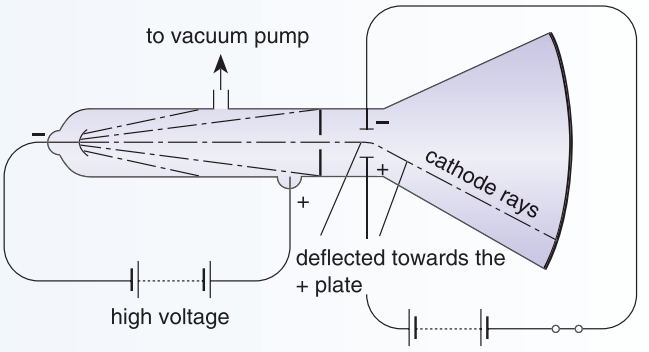


Fig. 5.23 Cathode rays are deflected towards the positive plate

Thomson suggested that the rays were composed of particles carrying negative charges. He called them electrons. Later Thomson discovered positive particles. He named them protons.

Thomson proposed that an atom should be thought of as being made up of negatively charged electrons moving around in a 'sea' of positive charge. This is commonly called the 'plum pudding' model (Fig. 5.24).

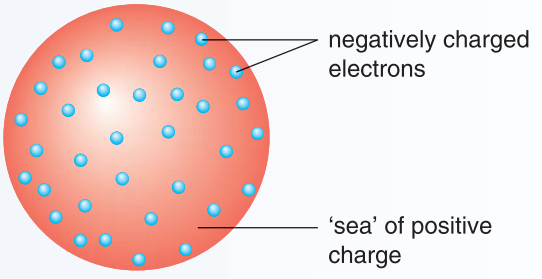


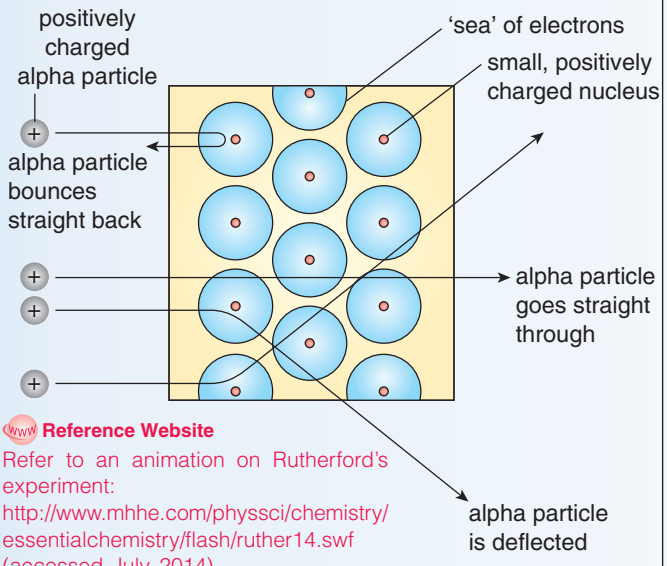
Fig. 5.24 The 'plum pudding' model

The nucleus — Rutherford's scattering experiment

In 1911, E. Rutherford carried out a number of crucial experiments.

Continued on next page

Thin sheets of gold foil were bombarded with positively charged particles called alpha particles. Most of the alpha particles went straight through the gold foils. Some were deflected. A few of the alpha particles bounced straight back (Fig. 5.25).



Reference Website
Refer to an animation on Rutherford's experiment:
http://www.mhhe.com/physsci/chemistry/essentialchemistry/flash/ruther14.swf
(accessed July 2014)

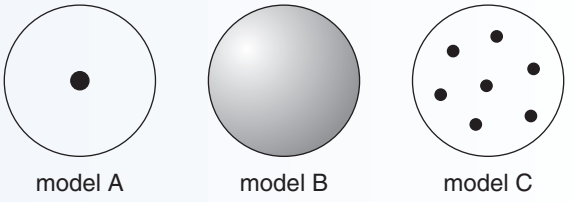
Fig. 5.25 Bombardment of gold foil with alpha particles

Rutherford concluded that most of the atom was empty space with a small, positively charged nucleus at its centre. This led to the downfall of the 'plum-pudding' model.

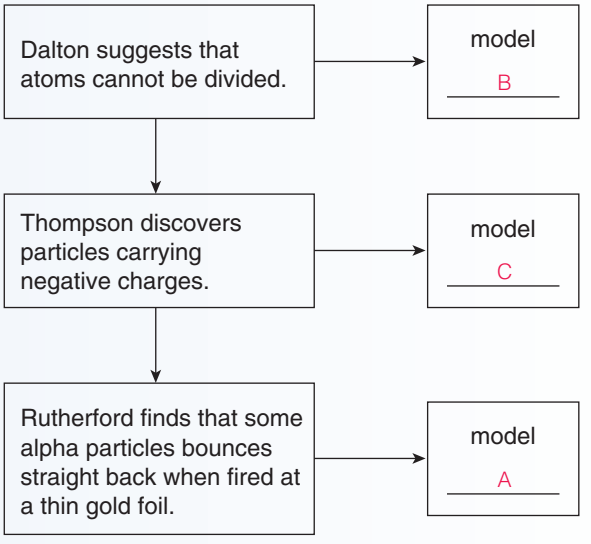
In 1932, J. Chadwick discovered uncharged particles, which he called neutrons. The proposals of Rutherford and Chadwick led to a model of the atom which was composed of protons, neutrons and electrons. Protons and neutrons were packed closely together in the nucleus. Electrons moved around the nucleus at a considerable distance.

Questions

1 The following diagrams show possible structures for atoms based on different models.



Complete the flow diagram by filling in the boxes with the letter corresponding to the appropriate model.



2 Why were neutrons more difficult to discover than electrons and protons? Neutrons are neutral particles and thus more difficult to detect than charged particles.

Have you mastered?

Key terms

atom 原子 2 element 元素 2 symbol 符號 3 metal 金屬 4 non-metal 非金屬 4

metalloid 類金屬 6 state 狀態 6 nucleus 原子核 7 proton 質子 7 neutron 中子 7

electron 電子 7 subatomic particle 次原子粒子 8 atomic number 原子序 8

mass number 質量數 9 isotope 同位素 12 relative isotopic mass 相對同位素質量 14

relative atomic mass 相對原子質量 15 relative abundance 相對豐度 15 orbit 軌道 17

shell 電子層 17 electronic arrangement 電子排佈 18 electron diagram 電子圖 18

orbital 軌態 21

Checklist

After studying this unit, you should be able to

- ☐ state the relationship between element and atom;
- ☐ use symbols to represent elements;
- ☐ classify elements as metals or non-metals on the basis of their properties;
- ☐ be aware that metalloids possess characteristics of both metals and non-metals;
- ☐ state and compare the relative charges and the relative masses of a proton, a neutron and an electron;
- ☐ describe the structure of an atom in terms of protons, neutrons and electrons;
- ☐ interpret chemical shorthand conventions such as $^{23}_{11}Na$;
- ☐ deduce the numbers of protons, neutrons and electrons in atoms with given atomic numbers and mass numbers;
- ☐ identify isotopes among elements with relevant information;

- ☐ perform calculations related to relative isotopic masses, relative atomic masses and relative abundance of isotopes;
- ☐ understand and deduce the electronic arrangements of atoms;
- ☐ represent the electronic arrangements of atoms using electron diagrams.

(Put a '✔' in the box if you have acquired the knowledge concerned.)

Summary

1 All elements are made of atoms.

2 Chemists use symbols to represent elements.

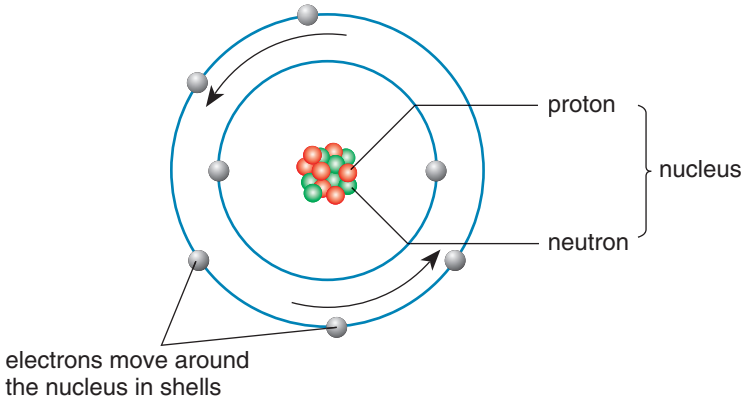
3 Elements can be classified into three main groups — metals, metalloids and non-metals.

4 a) An atom consists of three types of subatomic particles: protons, neutrons and electrons.

Name	Symbol	Relative charge	Relative mass
Proton	p	+1	1
Neutron	n	0	1
Electron	e^-	–1	$\frac{1}{1\,840}$

In a neutral atom, number of electrons = number of protons

b) The nucleus contains protons and neutrons. Electrons move around the nucleus in shells.



5 a) Atomic number of an element = number of protons in an atom of that element
= number of electrons in a neutral atom of that element

b) Only atoms of the same element have the same atomic number.

6 Mass number = number of protons + number of neutrons



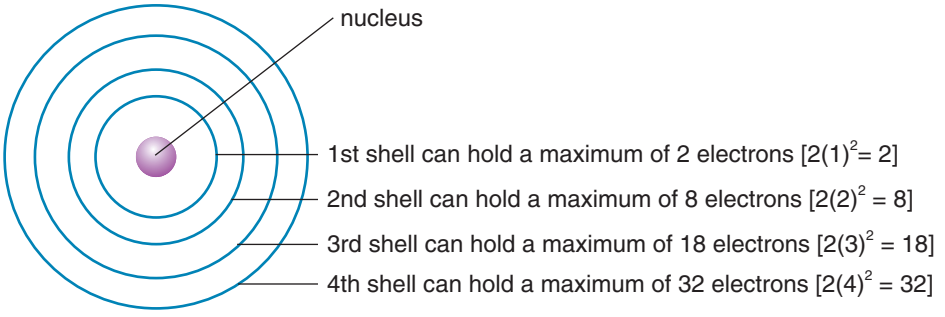
7 Isotopes are different atoms of an element which have the same number of protons but a different number of neutrons. For example, chlorine has two isotopes: Cl-35 and Cl-37.

8 a) The relative isotopic mass of a particular isotope of an element is the mass of one atom of the isotope compared with $\frac{1}{12}$ of the mass of one carbon-12 atom.

b) The relative isotopic mass of an isotope is roughly equal to its mass number.

9 The relative atomic mass of an element is the weighted average relative isotopic mass of all the naturally occurring isotopes of that element compared with $\frac{1}{12}$ of the mass of one carbon-12 atom.

10 The way in which electrons are arranged in an atom is called its electronic arrangement.



Have you mastered?

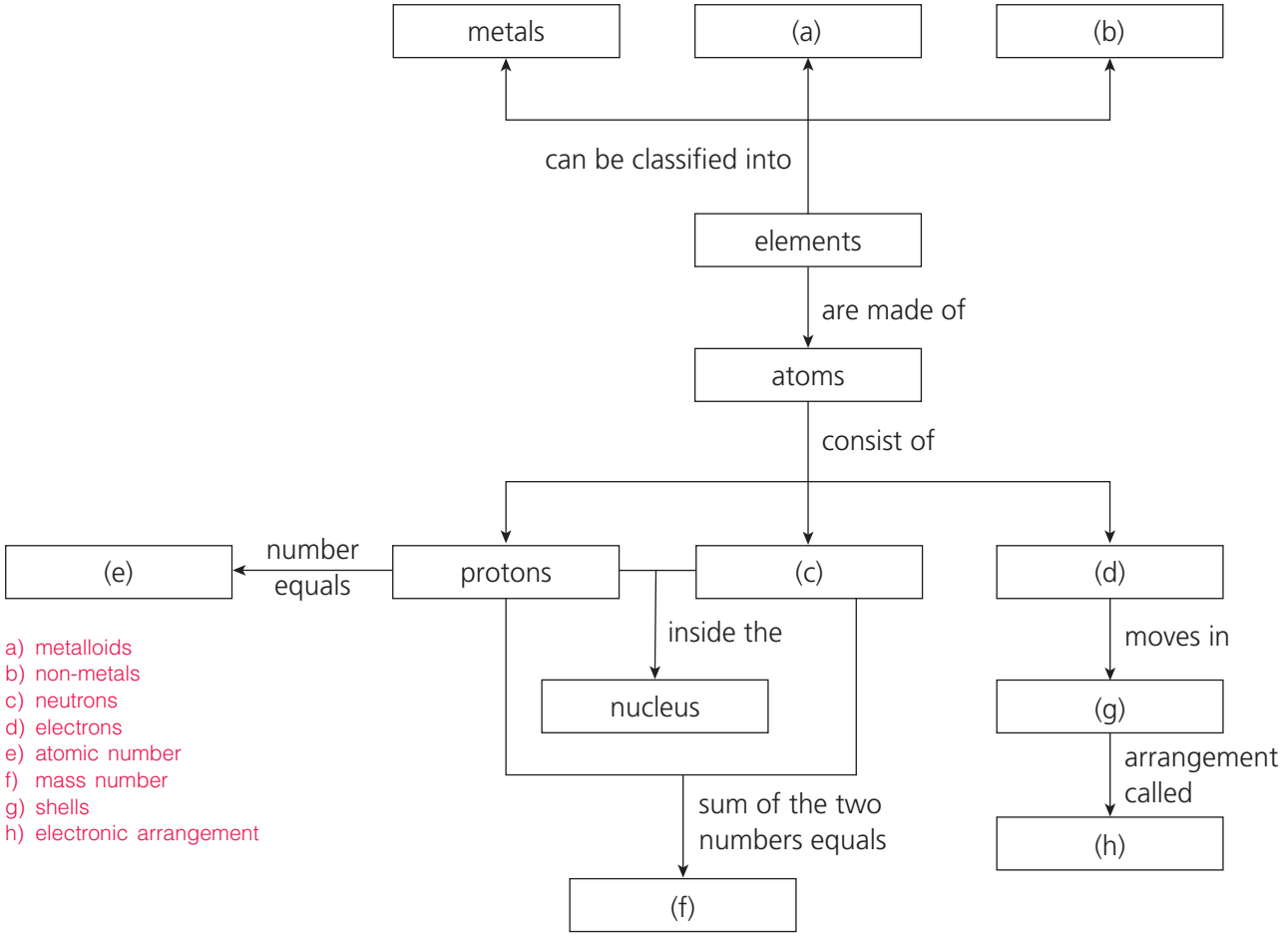
Unit Exercise

Note: The symbol ! indicates the level of difficulty of a question.

Part I Knowledge and understanding

1 Complete the concept map using the following words:

atomic number, electronic arrangement, electrons, mass number, metalloids, neutrons, non-metals, shells



a) metalloids
b) non-metals
c) neutrons
d) electrons
e) atomic number
f) mass number
g) shells
h) electronic arrangement

2 Decide whether each of the following statements is true or false. Explain your answer.

a) All atoms contain protons, electrons and neutrons. False
The most common isotope of hydrogen ($^{1}_{1}H$) does not contain any neutron.

b) One isotope of carbon has an atomic number of 6 and a mass number of 12. This means that it has twice as many neutrons as protons. False
The isotope has 6 protons and 6 neutrons.

c) Electron shells are negatively charged. False
Electrons that occupy the shells are negatively charged.

d) The second electron shell of an atom can hold a maximum of 8 electrons.
True
The maximum number of electrons the second electron shell can hold is $2(2)^2$, i.e. 8.

3 a) Complete the following table for atoms A to F. (The letters are NOT symbols of the atoms.)

Atom	Atomic number	Mass number	Number of			Electronic arrangement
			protons	neutrons	electrons	
A	3	6	3	3	3	2,1
B	8	18	8	10	8	2,6
C	11	23	11	12	11	2,8,1
D	16	32	16	16	16	2,8,6
E	8	16	8	8	8	2,6
F	13	27	13	14	13	2,8,3

b) Name the atoms A, B and C.
A — lithium B — oxygen C — sodium

c) Which of the above atoms are isotopes?
B and E

4 Give the names and symbols of the following elements (A to D) from the descriptions given.

Element	Description	Name	Symbol
A	Its atomic number is 7.	nitrogen	N
B	Its atom has 12 protons.	magnesium	Mg
C	Its atom has 18 electrons.	argon	Ar
D	The electronic arrangement of its atom is 2,8,8,1.	potassium	K

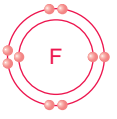
5 Draw electron diagrams of atoms of the following elements:

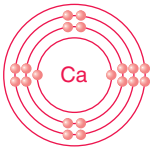
a) Fluorine

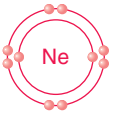
b) Calcium

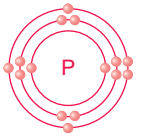
c) Neon

d) Phosphorus

a) 

b) 

c) 

d) 

7 Atomic number of an element
= number of protons in an atom of that element
∴ an atom of the element has 12 protons.
In a neutral atom, number of electrons = number of protons
∴ an atom of the element has 12 electrons.

11

Option	Mass number = number of protons + number of neutrons
A	12
B	18
C	19
D	40

Part II Multiple choice questions

6 The following table shows the properties of some unknown elements.

Property \ Element	X	Y	Z
State at room temperature and pressure	solid	solid	liquid
Appearance	dull	shiny	shiny
Electrical conductivity	nil	good	good

Which of the elements is / are metal(s)?

A X only
B Y only
C X and Z only
(D) Y and Z only

Metals are good conductors of electricity. They have a shiny appearance. Z is mercury.

7 The numbers of protons, neutrons and electrons in four different atoms, A, B, C and D, are shown below.

Refer to the previous page for explanation.

Which is an atom of an element that has an atomic number of 12?

	Number of protons in an atom	Number of neutrons in an atom	Number of electrons in an atom
A	4	4	4
B	6	6	6
C	6	12	6
(D)	12	12	12

(Edexcel GCSE (Higher Tier), Chemistry, Unit C1a, Jun. 2012, 31)

8 An atom has 25 electrons and 27 neutrons. Which of the following combinations concerning the atom is correct?

	Atomic number	Mass number
A	25	27
(B)	25	52
C	27	50
D	27	25

The atom has 25 electrons. It also has 25 protons. Thus, its atomic number is 25.

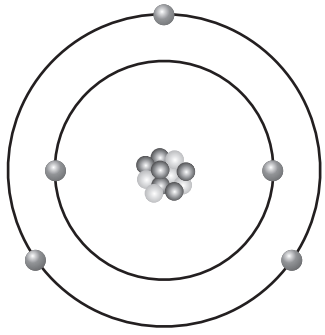
Mass number = number of protons + number of neutrons
= 25 + 27
= 52

9 An isotope of the element bohrium, $^{267}_{107}Bh$, has recently been discovered. How many neutrons are there in a nucleus of this isotope?

A 107
(B) 160
C 267
D 374

Number of neutrons
= mass number – atomic number
= 267 – 107
= 160

10 The diagram shows the structure of an atom of boron.



How many protons are in this atom?

(A) 5
B 6
C 11
D 16

In a neutral atom,
number of electrons = number of protons
The atom has 5 electrons. Thus, it also has 5 protons.

(Edexcel GCSE (Higher Tier), Chemistry, Unit C1a, Nov. 2011, 27)

11 The table below shows the numbers of protons, neutrons and electrons in four atoms, A, B, C and D.

Refer to the previous page for explanation.

Which atom has a mass number of 18?

	Number of protons in an atom	Number of neutrons in an atom	Number of electrons in an atom
A	6	6	6
(B)	8	10	8
C	9	10	9
D	18	22	18

(Edexcel GCSE (Foundation and Higher Tiers), Chemistry, Unit C2, Nov. 2011, 17)

12 Two atoms of sulphur have the symbols $^{32}_{16}S$ and $^{34}_{16}S$.

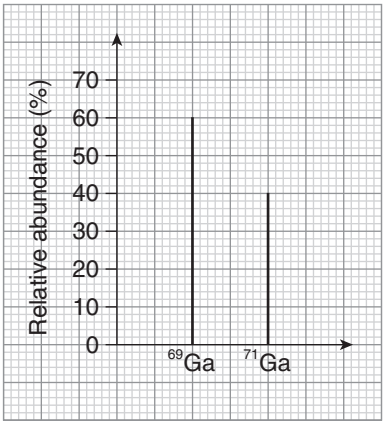
These two atoms will

$^{32}_{16}S$ contains 16 neutrons while $^{34}_{16}S$ contains 18 neutrons.

A react differently with oxygen under the same conditions.
B contain different numbers of electrons.
C have different atomic numbers.
(D) contain different numbers of neutrons.

(Edexcel GCSE (Foundation and Higher Tiers), Chemistry, Unit C2, Jun. 2012, 20)

13 Gallium has two isotopes, ^{69}Ga and ^{71}Ga. The graph below shows the relative abundance of the isotopes.



What is the relative atomic mass of gallium?

A 69.4
B 69.6
(C) 69.8
D 70.0

14 The relative atomic mass of boron is 10.8.

A sample of boron contains the isotopes boron-10 and boron-11.

What is the percentage abundance of each isotope in the sample?

	Percentage abundance of boron-10 (%)	Percentage abundance of boron-11 (%)
A	10	11
B	50	50
(C)	20	80
D	80	20

(Edexcel GCSE (Higher Tier), Chemistry, Unit C2, Jun. 2012, 40)

15 Which of the following elements are non-metals?

(1) C
(2) N
(3) S

A (1) and (2) only
B (1) and (3) only
C (2) and (3) only
(D) (1), (2) and (3)

Carbon (C), nitrogen (N) and sulphur (S) are non-metals.

16 Which of the following statements concerning the atom $^{31}_{15}X$ is / are correct?

(1) X is a non-metal.
(2) There are 16 electrons in an atom of X.
(3) There are 15 neutrons in an atom of X.

(A) (1) only
B (2) only
C (1) and (3) only
D (2) and (3) only

(1) X is phosphorus, a non-metal.
(2) The atomic number of X is 15. There are 15 electrons in an atom of X.
(3) Number of neutrons
= mass number – atomic number
= 31 – 15
= 16

17 An atom of element X has an electronic arrangement of 2,8,8,1. Which of the following statements concerning X is / are correct?

(1) X is a metal.
(2) Its mass number is 19.
(3) A compound of X gives a golden yellow flame in a flame test.

(A) (1) only
B (2) only
C (1) and (3) only
D (2) and (3) only

(1) X is potassium, a metal.
(2) Its atomic number is 19.
(3) A compound of X gives a lilac flame in a flame test.

13 Relative atomic mass of gallium $= \frac{69 \times 60 + 71 \times 40}{100}$
$= 69.8$

14 Let the relative abundance of boron-10 and boron-11 be y% and $(100 - y)$% respectively.

Relative atomic mass of boron $= 10.8 = \frac{10 \times y + 11 \times (100 - y)}{100}$
$1\,080 = 10y + 1\,100 - 11y$
$y = 20$

∴ the percentage abundance of boron-10 and boron-11 are 20% and 80% respectively.

Part III Structured questions

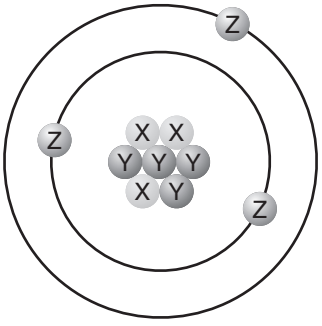
18 In 1807, Humphrey Davy discovered potassium by electrolyzing a molten compound of potassium. He found a shiny substance collected around the negative electrode. From the evidence he collected in experiments carried out on this substance, Davy decided that it was a metal.

Suggest TWO pieces of experimental evidence that Davy could have used to show that the shiny substance was a metal.

19 Give TWO differences in physical properties between a metal and a non-metal.

(Edexcel GCE O Level, Paper 1, May 2009, 14(a))

20 The diagram below represents the structure of an atom of isotope A of an element.



a) i) Name the particles X, Y and Z. X: proton Y: neutron Z: electron

ii) Which of the particles, X, Y and Z, have approximately the same mass?
X and Y

iii) Give the atomic number and mass number of this atom. Atomic number: 3 Mass number: 7

b) i) This element has another isotope, B, with a mass number one less than isotope A. State the composition of the nucleus of an atom of isotope B.
3 protons and 3 neutrons

ii) Why do these isotopes have the same chemical properties? They have the same number of electrons.

(Edexcel GCE O Level, Paper 1, Jan. 2009, 6(a)–(b))

21 This question is about atomic structure.

Look at the table. It shows some information about the particles which make up atoms.

Particle	Charge	Relative mass
Proton	positive	1
Electron	negative	0.0005
Neutron	neutral	1

a) Complete the table. b) i) Mass number of an atom is the sum of the numbers of protons and neutrons in the atom.

b) Look at the symbol for magnesium.

$^{24}_{12}Mg$

The mass number of magnesium is 24.

i) What is meant by 'mass number'?

ii) Write down the electronic arrangement of a magnesium atom.
2,8,2

iii) A magnesium atom is electrically neutral. Explain why. It has the same number of protons as electrons.

(OCR GCSE Gateway Science, Chem. B, Unit 1, Jan. 2011, 10)

22 All naturally occurring sodium atoms have a relative isotopic mass of 23. The atoms are represented by the symbol ^{23}Na. However, radioactive isotopes of sodium, e.g. ^{24}Na, may be prepared. Refer to page T20.

a) State the number of protons, neutrons and electrons in an atom of ^{23}Na.

b) Draw an electron diagram for an atom of ^{23}Na.

c) Define the term 'relative atomic mass'.

d) A sample of sodium from a nuclear reactor contains 7.00% of ^{24}Na and 93.0% of ^{23}Na by mass. Calculate the relative atomic mass of the sample.
23.07

18 Any two of the following:
- It is a good conductor of electricity.
- It is a good conductor of heat.
- It is malleable.
- It is ductile.
- It reacts with water to give hydrogen.

19 Any two of the following:
- Metal conducts electricity but non-metal does not.
- Metal conducts heat well but non-metal does not.
- Metal has a high melting point (or boiling point) but non-metal has a low melting point (or boiling point).
- Metal is shiny but solid non-metal is dull.
- Metal is malleable / ductile but solid non-metal is brittle.

23 Europium, atomic number 63, is used in some television screens to highlight colours. A chemist analyzed a sample of europium. The results are shown in the table below. Refer to page T20.

Isotope	Relative isotopic mass	Relative abundance (%)
^{151}Eu	151.0	47.77
^{153}Eu	153.0	52.23

a) Define the term 'relative isotopic mass'.

b) Using the above table, calculate the relative atomic mass of the europium sample, giving your answer to TWO decimal places. 152.04

c) Isotopes of europium have differences and similarities.

i) In terms of protons, neutrons and electrons, how is an atom of ^{151}Eu DIFFERENT from an atom of ^{153}Eu?

ii) In terms of protons, neutrons and electrons, how is an atom of ^{151}Eu SIMILAR to an atom of ^{153}Eu?

(OCR Advanced Subsidiary GCE, Chem. A, F321, Jan. 2010, 1)

24 A sample of copper contains two isotopes. Refer to page T21.

a) What are 'isotopes'?

b) i) Complete the table below for the two isotopes of copper.

Isotope	Atomic number	Mass number	Number of protons	Number of neutrons	Number of electrons
1	29	63	29	34	29
2	29	65	29	36	29

ii) The relative atomic mass of this sample of copper is 63.6. What is the relative abundance of isotope 1? 70.0%

c) Explain why the two isotopes of copper have identical chemical properties.

25 The current model of the atom describes a nucleus containing protons and neutrons surrounded by electrons. Refer to page T21.

a) Complete the following table showing the properties of the subatomic particles.

Subatomic particle	Relative mass	Relative charge
Proton	1	+1
Neutron	1	0
Electron	negligible	–1

b) Explain why

i) mass numbers of atoms are always whole numbers;

ii) the relative atomic mass of an element may NOT be a whole number.

(OCR Advanced Subsidiary GCE, Chem. B (Salters), F331, Jan. 2009, 1(b))

26 The melting points and boiling points of four elements at 1 atmosphere are listed in the table below.

Elements	Melting point (°C)	Boiling point (°C)
Argon	–189	–186
Bromine	–7	59
Chlorine	–101	–35
Sulphur	119	445

Refer to page T21.

a) Which of the above substances exist as a liquid at –90 °C and 1 atmosphere?

b) The four substances are non-metals. Write FOUR typical properties, besides melting and boiling points, of non-metals.

c) Draw electron diagrams for atoms of the following elements:

i) argon;

ii) chlorine;

iii) sulphur.